

Compulsory part:

Fill all cells of the following table conform to the IP address calculation rules!

IPv4 address	Network connected directly to two other networks	Address Mask, Prefix	Address of the 1 st host in the network	Address of the last host in the network	Broadcast address	No. of addresses usable for any node interfaces
13.13.128.41	Yes		13.13.128.33	13.13.128.61		
1.0.34.10	No		1.0.34.1		1.0.34.255	
127.0.0.1	Yes	192.0.0.0				
X	No	/27			13.12.5.159	
66.12.14.2	Yes		66.12.14.1		66.12.14.3	
X	Yes	/19		13.6.227.253		
Supernet of: 111.172.0.0/16 111.173.0.0/.....	No					131070
Supernet of: 114.10.2.0/27 114.10.2.32/28 114.10.2.48/.....	Yes	/26				

II. Theory part:

1. Explain in detail following mechanisms: aggregation, fragmentation, modulation, attenuation, amplification and encapsulation (text and figures are required)!
2. Explain elements and communication functions of the Hybrid reference models (text and figures are required)!
3. Suppose that you have a communication channel (Channel 1) with $\frac{P_1}{P_0} = 1/2$ attenuation. Knowing that $\log_{10} 2 = 0.3$ and $80 \approx 81$ (with error level 2%) determine amplification in dB of another channel (Channel 2) having ratio of powers $\frac{P_2}{P_0} = 140$!
4. Compare and explain the IEEE 802.3 and IEEE 802.5 communication mechanisms (text and figure is required)!
5. Describe in detail the addressing modes of the IPv4/IPv6 communication mechanisms in function of the source and destination addresses (text and figures are required).
6. Explain the protocol data unit of the UDP (text and figure is required)!

Computer Network Architectures and Protocols
Exam questions

December 18th, 2025, 18:00

I. Compulsory part:

Fill all cells of the following table conform to the IP address calculation rules!

One of the IPv4 addresses used in the network	Number of 1-hop neighbor networks	Address Mask, Prefix (in both formats)	Address of the 1 st interface in the network	Address of the last end node interface in the network	Broadcast address	No. of addresses usable for any node interfaces
202.133.128.41	2	127 255.255.255.224	202.133.128.33	202.133.128.60	202.133.128.63	28
151.135.117.1	0	122 255.255.252.0	151.135.116.1	151.135.119.254	151.135.119.255	$32-22$ 2 = 1024
127.13.0.4	3	255.224.0.0 199	127.0.0.1	127.31.255.251	127.31.255.255	$32-11$ 2 - 2 - 3
Not specified	7	/8 255.0.0.0	107.0.0.1	107.255.255.247	107.255.255.255	$32-8$ 2 - 2 - 7
192.168.7.1	0	130 255.255.255.252	192.168.7.1	192.168.7.2	192.168.7.3	2
Not specified	1	/18 255.255.192.0	5.6.64.1	5.6.127.253	5.6.127.255	$32-18$ 2 - 2 - 1
Supernet of: 101.18.0.0/16 101.19.0.0/16	4	115 255.254.0.0	101.18.0.1	101.19.255.250	101.19.255.255	131068
Supernet of: 14.168.2.0/24 14.168.3.0/25 14.168.3.128/25	8	/23 255.255.254.0	14.168.2.1	14.168.3.246	14.168.3.255	$32-23$ 2 - 2 - 8

II. Theory part:

1. Explain in detail following mechanisms: aggregation, fragmentation, modulation (text and figures are required)!
2. Explain elements and communication functions of the OSI reference model (text and figures are required)!
3. Knowing that the attenuation (A) is 3 dBm (with 2% accuracy) for ratio $P_1/P_0 = 2$, calculate and prove without calculator the value of A in dB for ratio $P_2/P_0 = 210$ with 2% accuracy! How much is P_2 in Watts?
4. Compare medium access mechanism of the ALOHA technology variants and determine by calculations the maximum throughput of them (text, figures and formulae are required)!
5. Describe in detail the packet header structure of the IPv4 and IPv6 communication mechanisms (text and figures are required)!
6. Explain the protocol data unit of the UDP mechanism (text and figure is required)!

Name: _____

Neptun Code: _____

Signature: _____

Computer Network Architectures and Protocols*Exam questions*June 14th, 2022 14:00**I. Compulsory part:**

Fill all free cells and "....." fields of the following table conform to the IP address calculation rules!

	IPv4 address	Network connected to any other network	Address Mask, Prefix	Address of the 1 st host in the network	Address of the last host in the network	Broadcast address	No. of addresses usable for end node interfaces
1.	110.109.108.107	Yes	255.255.254.0				
2.	110.109.108.107	No	/17				
3.	127.0.0.1	Yes	128.0.0.0				
4.	X	No	/30			101.128.5.67	
5.	11.11.10.10	Yes		11.11.10.9		11.11.10.15	
6.	X	No	/27		193.6.0.30		
7.	Supernet of: 112.110.0.0/16 112.111.0.0/.....	Yes					2^{17-4}
8.	Supernet of: 12.18.4.0/24 12.18.5.0/25 12.18.5.128/.....	No	/23				510

II. Theory part:

1. Explain following mechanisms: aggregation, fragmentation, modulation, attenuation, amplification and encapsulation (text and figures are required)!
2. Explain the layered network communication models (text and figure are required)!
3. Explain in detail the medium access and control mechanism of the IEEE 802.3 technology (text and figure are required)!
4. Compare and explain the versions of the ALOHA communication mechanism (text and figure is required)!
5. Explain simultaneous CDMA communication mechanisms of data A_1 , B_1 , C_0 , if the symbol codes used are:
 $A_1 = (1, 1, 1, 1, 1, 1, 1, 1)$; $B_1 = (1, 1, -1, -1, 1, 1, -1, -1)$; $C_1 = (1, 1, 1, 1, -1, -1, -1, -1)$!
6. Classify and list elements of the routing protocol types used in the Internet (text is required)!
7. Explain the protocol data unit of the TCP (text and figure is required)!
8. Knowing that the attenuation (A) is 3 dBm for ratio $P_1/P_0 = 2$, calculate and prove without calculator the value of A in dB for ratio $P_2/P_0 = 210$! How much is P_2 in Watts?
Indication: can be used approximation $49 \sim 50$.

I. Compulsory part:

Fill all cells of the following table conform to the IP address calculation rules!

IPv4 address	Network connected to any other network	Address Mask, Prefix	Address of the 1 st host in the network	Address of the last host in the network	Broadcast address	No. of address usable for end node interface
193.6.128.5	Yes	255.255.0.0				
135.149.129.129	No	/17				
127.0.0.1	Yes	128.0.0.0				
X	No	/30			135.128.5.193	
2.2.2.1	Yes		2.2.2.1		2.2.2.3	
X	No	/19		193.6.0.30		
Supernet of: 172.18.0.0/16 172.19.0.0/____	No					131070
Supernet of: 192.168.0.0/24 192.168.1.0/25 192.168.1.128/____	Yes	/23				

II. Theory part:

1. Explain in detail following mechanisms: aggregation, fragmentation, modulation, attenuation, amplification and encapsulation (text and figures are required)!
2. Explain in detail the elements and mechanism of the general communication functions (text and figure are required)!
3. Explain in detail the medium access and control mechanism of the IEEE 802.3 technology (text is required)!
4. Describe the intermediate node types and their service production in the three lower layers of the communication architecture (text and figure are required)!
5. Compare and explain the versions of the ALOHA communication mechanism (text and figure are required)!
6. Classify and list elements of the routing protocol types used in the Internet (text is required)!
7. Explain the protocol data unit of the UDP (text and figure is required)!
8. Give five application layer protocols and explain the essence of the service provided (text is required)!